

COMBUSTION CHEMISTRY ASSIGNMENT

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1 Exercise - 01

Calculation of air-fuel ratio, mole fraction and mixture stoichiometry

Exercise - 01 has been coded using MATLAB instead of Python.

NOTE:

The code has been written such that the function runs with or without the user input.

The program is written such that you can specify the hydrocarbon by entering (x,y) value of C_xH_y and also the you can provide the amount of hydrocarbon available to calculate the moles of air required.

Three different ways to run the MATLAB code.

1st code: tutorial1

Here the code assumes the value of x and y as 1 and 4 creating Methane and the amount of Methane available is 55.5 litres.

2nd code: tutorial1(x,y)

Here you can specify the hydrocarbon by providing the value of x and y while the amount of hydrocarbon available is assumed to be 55.5 litres to calculate the amount of oxidiser required.

3rd code: tutorial1(x,y,z)

Here you can specify the hydrocarbon by providing the value of x and y and the amount of hydrocarbon available in litres.

2 Exercise - 02

Calculation of heating values and adiabatic flame temperature for hydrocarbon fuels

Exercise -02 is solved in Python.

Results

Behaviour of Adiabatic Flame Temperature with different equivalence ratio. This behaviour can be explained in two different regime (lean mixture ($\phi < 1$) and rich mixture($\phi > 1$))

As ϕ increases, the adiabatic flame temperature value increases up to $\phi = 1$ and then again the AFT decreases if the ϕ is increased.

This is because in lean mixture excess amount of oxidiser available will absorb the heat generated from combustion leading to decrease in AFT, hence we can see that as ϕ value increases the amount of oxidiser excess available decreases hence less heat is lost hence higher value of AFT.

For rich mixture the not all the fuel available is burnt and the partially burnt fuel doesn't contribute to overall increase in AFT. Furthermore, partial combustion of fuel leads to formation of compounds such as CO which absorbs the heat decreasing the AFT.

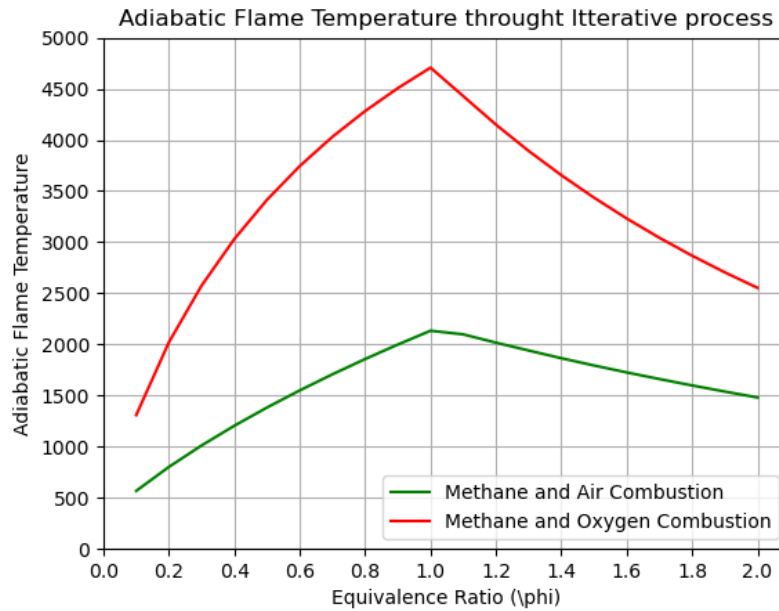


Figure 1: Adiabatic Flame temperature for methane combustion with air and oxygen at different Equivalence Ratio.

Adiabatic flame temperature can be calculated using sensible enthalpy, know the enthalpy of formation. It is given by the equation

$$h_k = \Delta h_f^0 + \int_{T_0}^{T_{AFT}} C_{pk} dT$$

[3]

$$H_{product}|_{298K} - H_{reactants}|_{298K} = \sum C_p|_{products}(T_{AFT} - T_{ref})$$

From the equation we can say that, the AFT depends on enthalpy of formation. Hence when CH_4

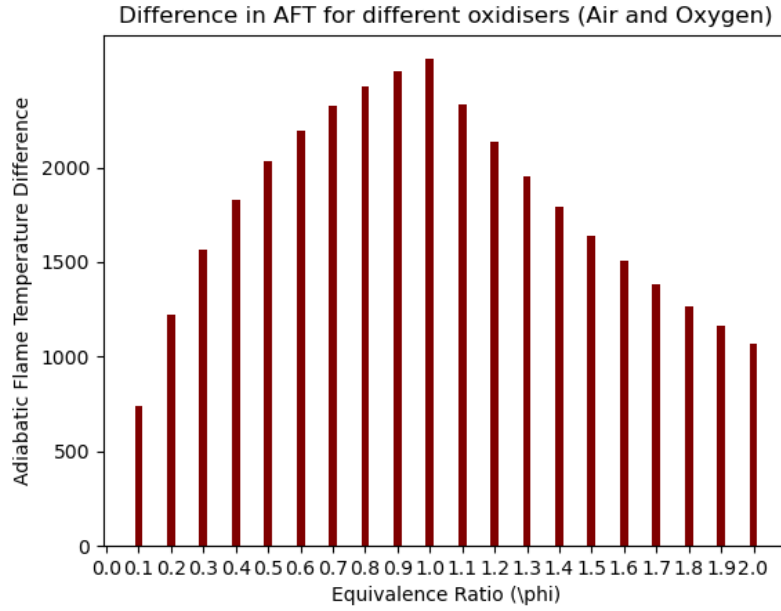


Figure 2: Adiabatic Flame temperature difference at each equivalence ratio for methane combustion with air and oxygen (Exercise 2).

is combusted with air other products such as N_2 , NO_x is formed. More the products, less the AFT, hence we can say that for oxygen as oxidiser we get more AFT than air as oxidiser.

3 Exercise - 03

Determination of adiabatic flame temperature for methane using CANTERA

Exercise - 03 was solved in Python using CANTERA and AFT obtained from 2nd exercise was imported into the present exercise to compare the results between iterative procedure and results obtained from CANTERA.

Result:

The one of the important difference between both the problems is the initial temperature. The iterative procedure was calculated base on $T_0 = 298K$ where as while using CANTERA $T_0 = 300K$.

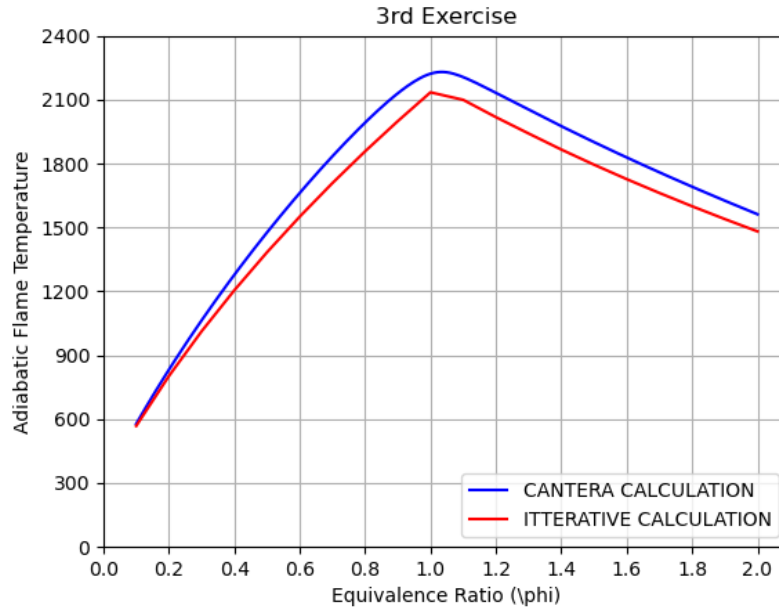


Figure 3: Comparison of adiabatic flame temperature for methane air combustion calculated by iterative process and using CANTERA.

$$H_{product}|_{298K} - H_{reactants}|_{298K} = \sum C_p|_{products}(T_{AFT} - T_{ref})$$

As per this formula for the calculation of AFT, the initial condition considered has a slight influence.

Not only the initial condition is important in the above equation but also important for finding the stoichiometric coefficient for fuel rich composition where equilibrium constant is used to calculate all the coefficients.

$$K_p = 10^{-2.4198 + 0.0003855T + \frac{2180.6}{T}}$$

Since there is a difference in the AFT that also affects the K_p value which is used to calculate the stoichiometric coefficients in the iterative process. Hence the deviation between the AFT values obtained from both the procedures shows a noticeable difference in the rich mixture regime.

4 Exercise - 04

Prediction of NO concentration at varying equivalence ratio

The exercise was coded using Python and the AFT from exercise 2 and 3 are imported for calculating $[NO]$.

Results: Concentration of $[NO]$ while $[N_2]$ and $[O_2]$ is calculated for the formula given below. (Zel-dovich Mechanism)

$$[NO] = 2K_{1f} \sqrt{\frac{K_p P_0}{R_u T_{AFT}}} [N_2][O_2]^{1/2}$$

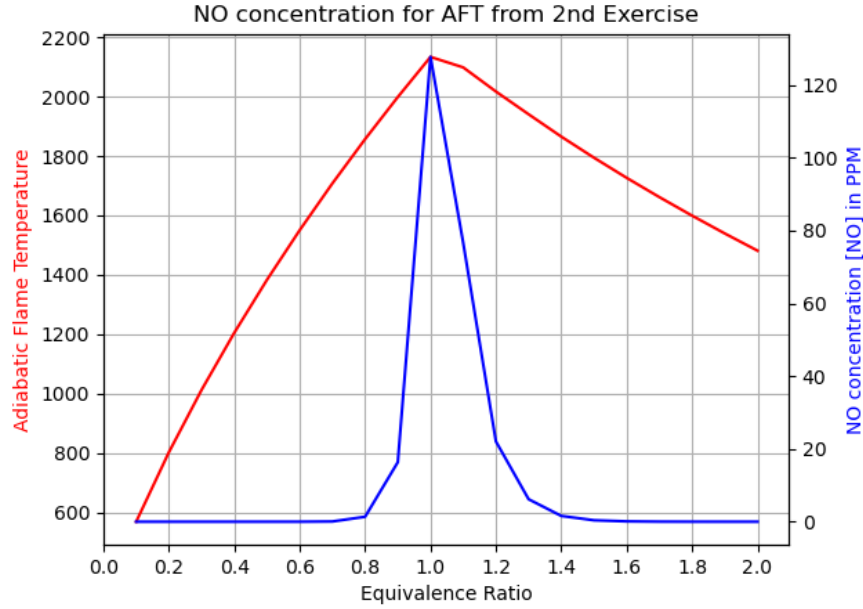


Figure 4: NO concentration obtained for the AFT calculated in exercise - 2.

It can be observed from figure 4 and figure 5 that the maximum change in $[NO]$ is observed between the equivalence ratio $0.8 < \phi < 1.4$

This behavior can be explained with respect to forward rate of reaction. $k_{1f} \propto e^{-\frac{1}{T_{AFT}}}$

ϕ	Adiabatic Flame Temperature (T_{AFT})	Forward rate (K_{f1})
0.8	1857.11K	0.03384357080955953
0.9	1998.9K	0.20446726669064516
1.0	2.134.03K	0.8812676610713742
1.1	2098.76K	2.9617960349533887
1.2	2017.5K	2.1903276452046945
1.3	1940.83K	1.0512869879278395
1.4	1.8354K	0.4971511871614046

The $[NO]$ increases exponentially with increase in AFT. Hence it can be observed that after a certain AFT there is an exponential increase in $[NO]$ and then as AFT decreases the $[NO]$ also decreases.

From the table it can be observed that with increase in 0.1 ϕ the K_{f1} value increases around 10 times. and then starts to decrease as AFT for rich composition decreases after $\phi = 1$. Hence we see a exponential increase and exponential decrease over a small range of ϕ .

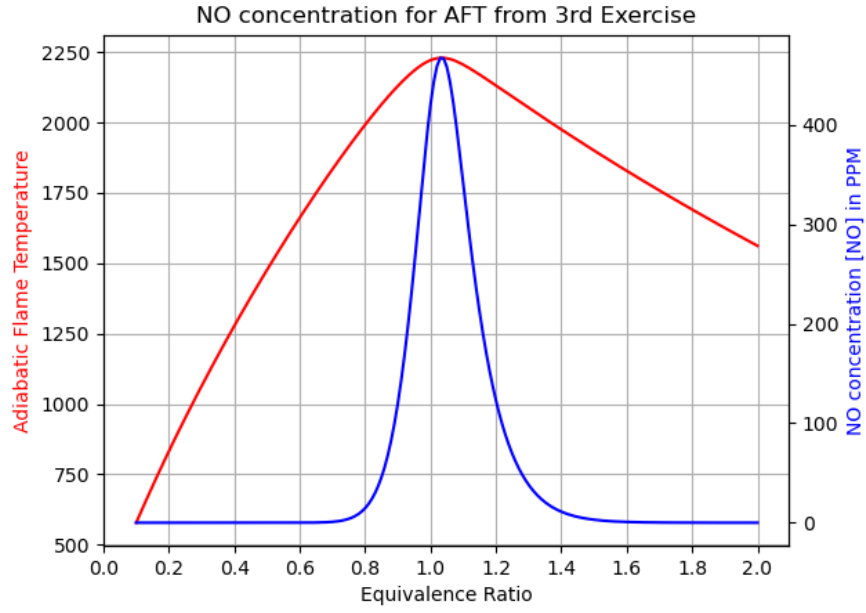


Figure 5: NO concentration obtained for the AFT calculated in exercise - 3

From Figure 4 and Figure 5 we can see that a small change in Adiabatic flame temperature around $\phi = 1$ can lead to a significant increase in the $[NO]$ concentration. This is because of the exponential growth of $[NO]$ concentration above 1800K. [4]

5 Exercise - 05

Determination of laminar flame speed of premixed methane-air mixture

Exercise - 05 was solved in Python using CANTERA.

Results:

- Variation of Adiabatic flame speed for different equivalence ratio.

$$AFS \propto \sqrt{K_p}$$

and we know that K_p depends on T_{AFT} .

Hence we can see that, the variation of laminar flame speed with respect to equivalence ratio has same behavior as variation of Adiabatic flame temperature with respect to equivalence ratio.

The maximum flame speed is obtained slightly toward rich mixture ($\phi = 1.05 - 1.07$) because, the flame speed is also influenced by H^+ and OH^- radical formation. Hence the peak is observed at a point where both radicals are produced and also the flame temperature is considerably high.

Effect of Pressure on laminar Flame speed

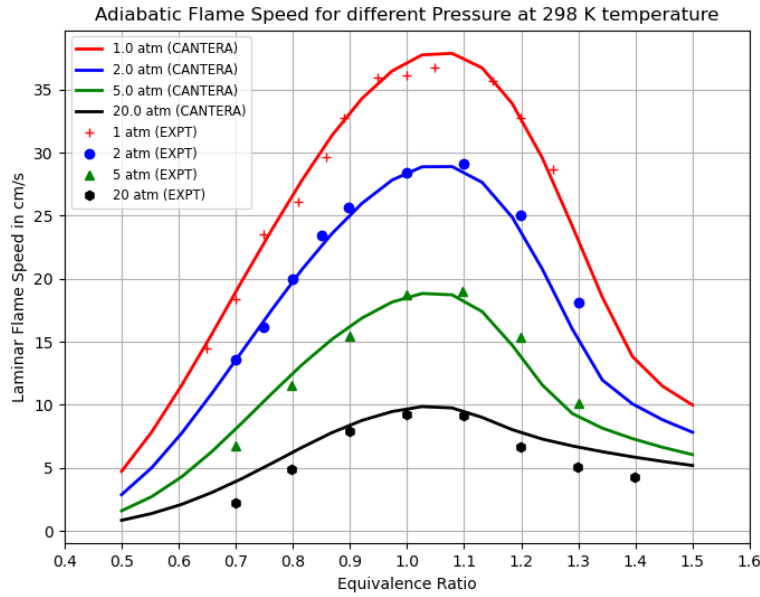


Figure 6: Laminar Flame Speed versus Equivalence Ratio for different initial Pressure and at constant initial temperature of 298K.

Figure 6 shows the variation of laminar flame speed for different equivalence ratio at different pressure. As the pressure increases the flame speed decreases. This behaviour is because of the influence of reaction rate on flame speed.

From Modified Arrhenius Equation

$$K_p = AP^n \exp \frac{-E_A}{RT}$$

here n is reaction order with respect to pressure. ($n < 2$ — — — — — From experiments)

From laminar flame theory we can say that,

$$LFS \propto \sqrt{\alpha K_p / \rho}$$

and $\alpha \propto \frac{1}{P}$ and $K_p is \propto P^n$
hence [2],

$$LFS \propto P^{\frac{n-2}{2}}$$

From this we can say that

$$LFS \propto \frac{1}{P^m}$$

From this we can say that as Pressure increases the LFS decreases.

Effect of Pressure on laminar Flame speed

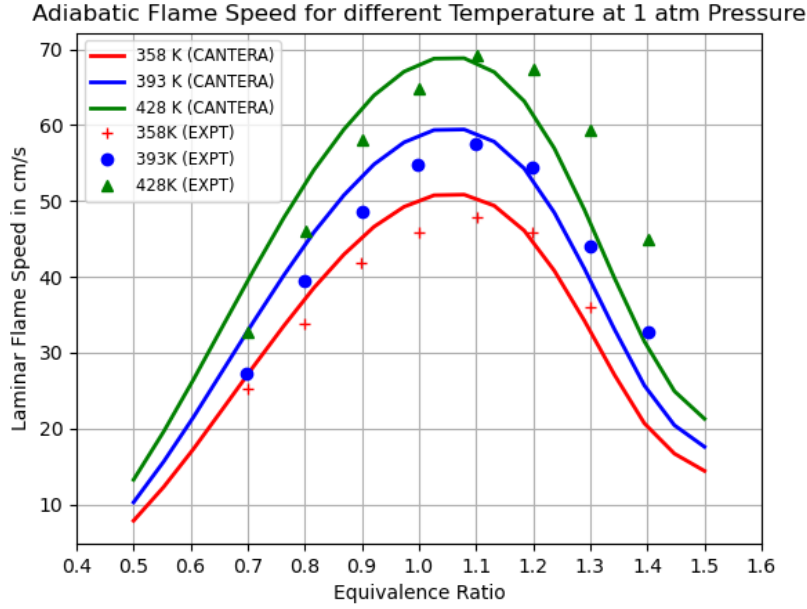


Figure 7: Laminar Flame Speed versus Equivalence Ratio for different initial temperatures and at constant initial Pressure of 1atm

Similar to pressure the initial temperature also affects the Laminar flame speed. From Laminar flame theory it has been obtained that,

$$LFS \propto T^n$$

here $n = 1.8$ for methane - air mixture (Obtained from experimentation).

From this we can say that as the initial temperature increases the LFS also increases.

6 Exercise - 06

Determination of ignition delay time of methane-air mixture using Cantera

Exercise - 06 was solved in Python using CANTERA.

Results

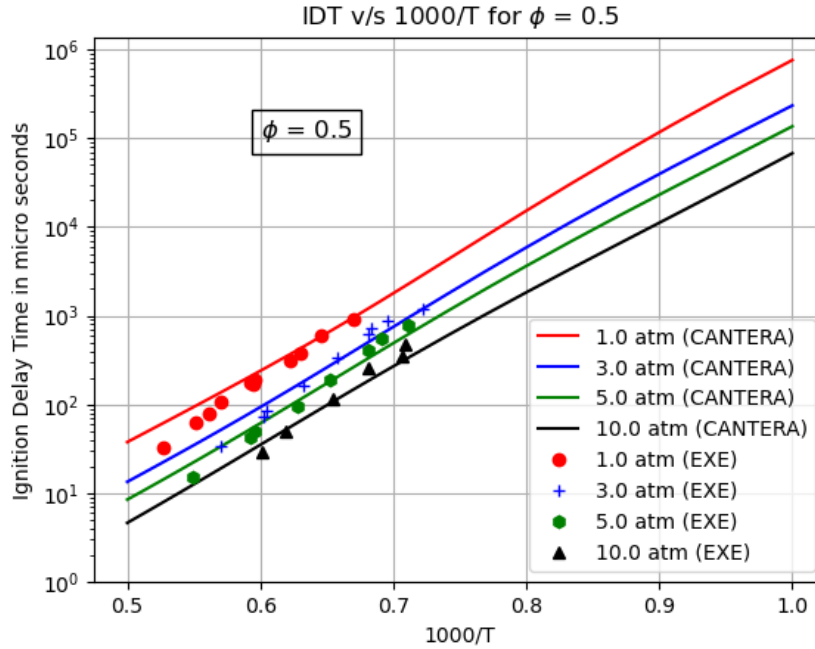


Figure 8: Variation of ignition delay time with temperature for different pressure values at $\phi = 0.5$

Variation of Ignition flame delay with change in equivalence ratio ϕ

For a given temperature and pressure the ignition delay time variation can be observed in all there regime.

1. **Lean Mixture:** Due to low concentration of fuel the time required to release the energy that is required for ignition increases.
2. **Equivalence Ratio = 1:** Here the amount of energy released tends to be maximum as there is no excess air or fuel to absorb the energy released and leads to ignition in shortest time possible.
3. **Rich Mixture:** Here the amount of energy released is high and hence the ignition delay is maintained to be low but surplus increase in fuel can also absorb the energy released leading to increased ignition delay time.

NOTE: For the Figure 8, Figure 9 and Figure 10 it can be observed that the variation in ignition delay time is not much but considering it to be draw on semi-log graph the smallest change can be by a factor of 10.

Variation of Ignition flame delay with change in pressure at a given equivalence ratio

Increase in pressure leads to increase in collision frequency by reducing mean free path leading to increased reaction rate and increased energy release required for ignition. Hence it can be observed that with increase in pressure the ignition delay time is decreased for all the three cases ($\phi = 0.5, 1.0, 1.5$)

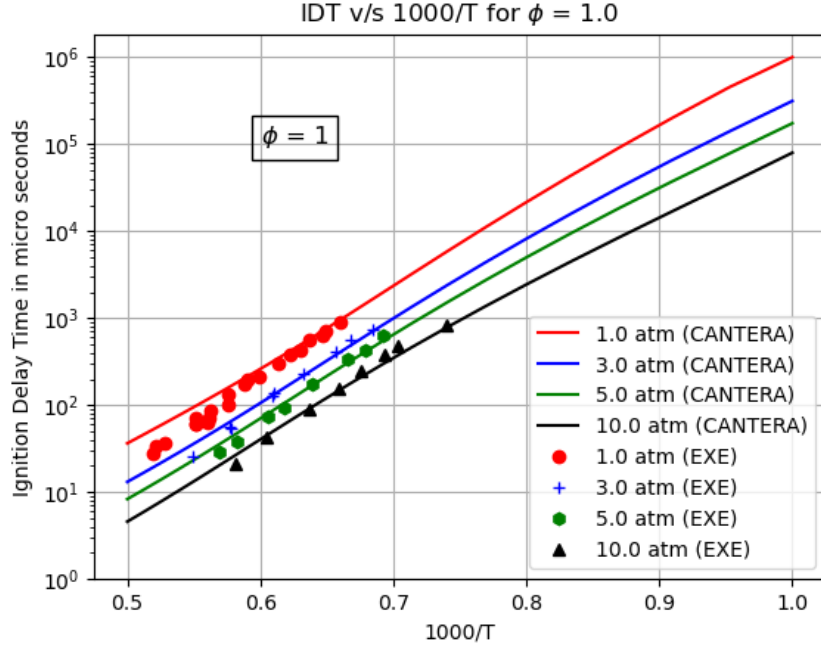


Figure 9: Variation of ignition delay time with temperature for different pressure values at $\phi = 1$

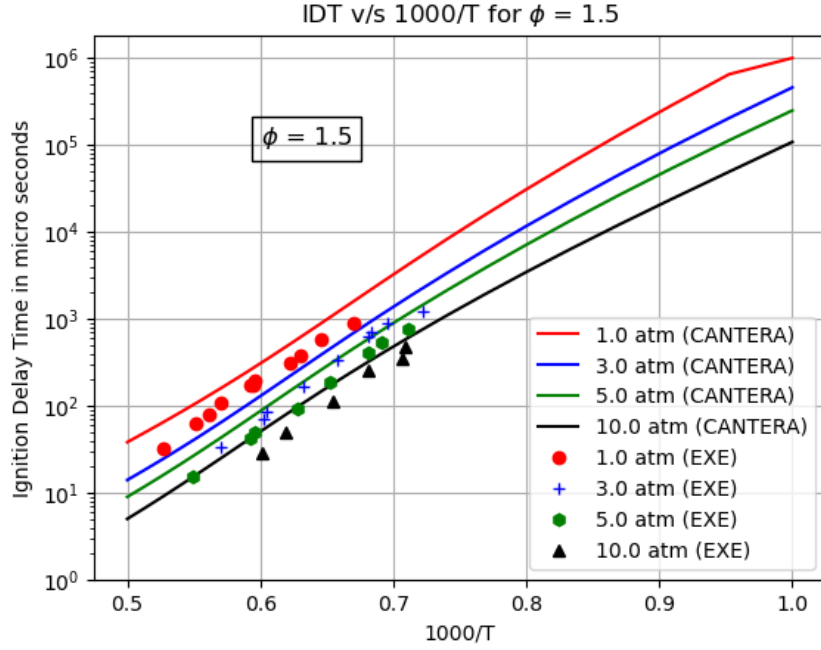


Figure 10: Variation of ignition delay time with temperature for different pressure values at $\phi = 1.5$

Application: The compression ratio of diesel Engine is much higher compared to petrol engine and during normal operation the equivalence ratio of both engine is close to each other and hence the ignition delay time of diesel engine is smaller compared to petrol engine

7 Exercise 7: Sensitivity and path flux analysis for methane-air combustion in Cantera

Results:

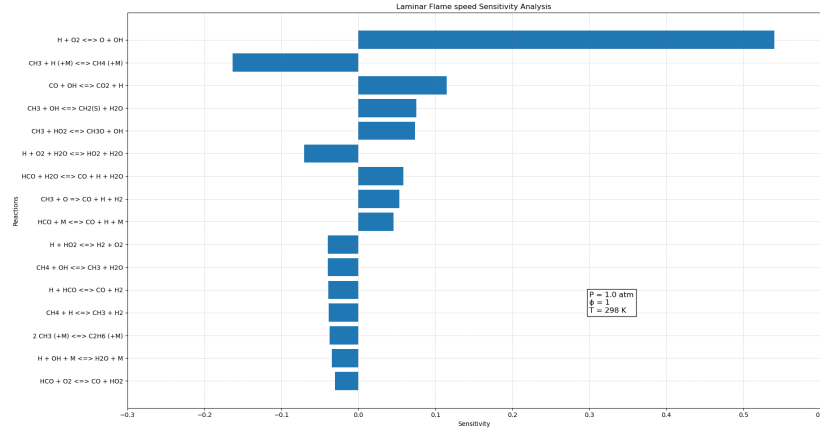


Figure 11: Laminar Flame Speed Sensitivity analysis at 1 atm Pressure

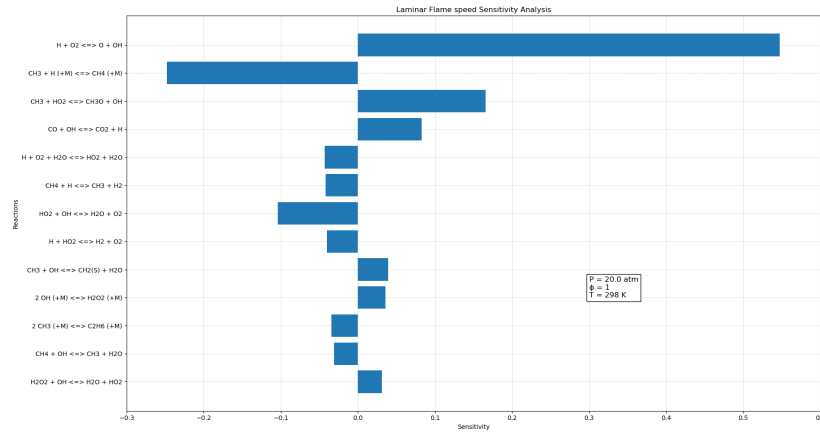
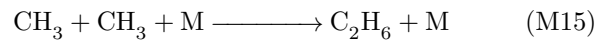


Figure 12: Laminar Flame Speed Sensitivity analysis at 20 atm Pressure

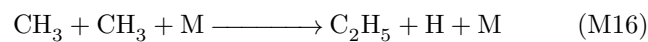
With increase in pressure the concentration of third body increases leading to change in the reaction pathway. We know that there are two pathways that depending on the condition,



At high pressure (M1) reaction is dominated because of the increase in concentration of M. while at low pressure (M2) is more favorable as (M1) requires large activation energy. Furthermore increase in concentration of M can cause Methyl radical self reaction



or



(M15) is chain termination reaction which becomes the main reaction pathway at high pressure leading to different sensitivity values for the equation.

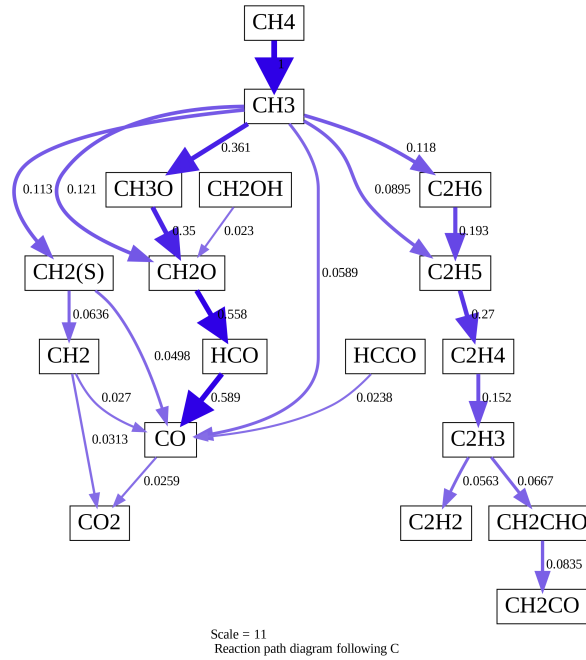


Figure 13: Reaction Path Diagram for 1 atm Pressure

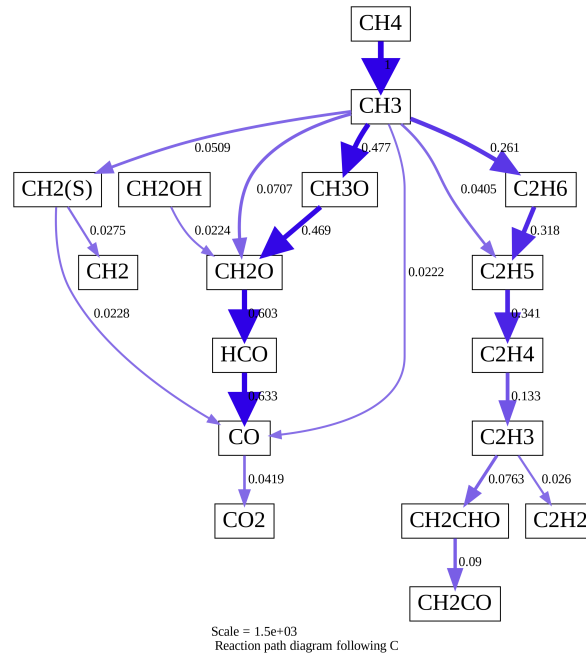


Figure 14: Reaction Path Diagram for 20 atm Pressure

For the reaction path diagram we can see that the main reaction for low pressure is dominated by (M2) leading to formation of CH_3O , CH_2O , CH_0 , CO and CO_2 .

While for high Pressure it can be observed that self reaction of Methyl radical is dominate and the that is the main reaction path

8 Exercise 8:

Development of a mechanism for a methane-air mixture using Reaction Mechanism Generator (RMG)

Results:

It can be observed that the Laminar Flame speed calculated by RMG shows an increased velocity

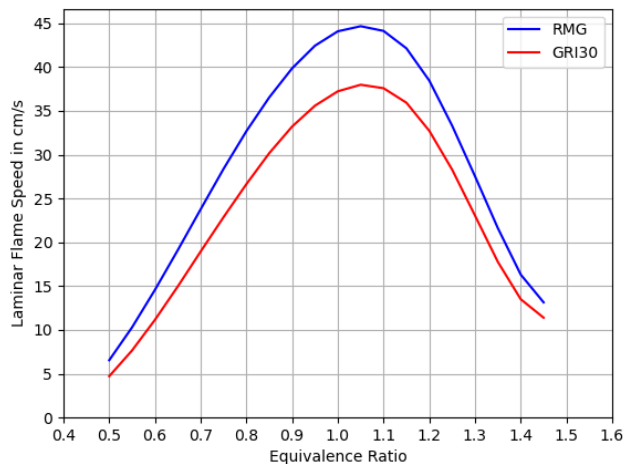


Figure 15: Laminar Flame Speed comparison between GRI3.0 mechanism and RMG generated mechanism

through out the range compared to GRI3.0. The accuracy can be improved by adding additional intermediate species that were left out. Apart from that by including more species on the edge can improve the accuracy but the simulation time increases.

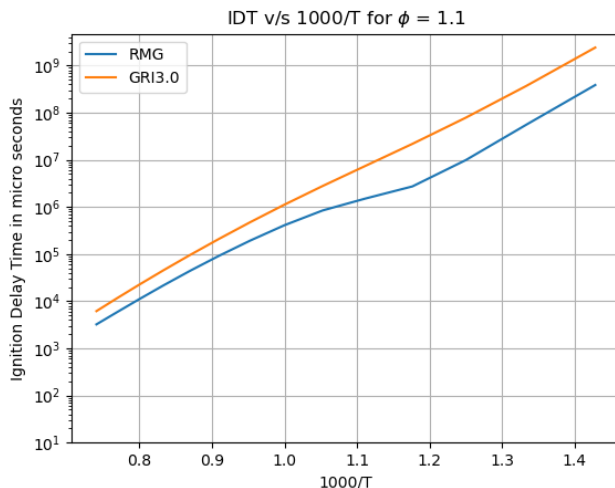


Figure 16: Ignition Delay time comparison between GRI3.0 mechanism and RMG generated mechanism

For temperature lower than 1000K it can be observed that the ignition delay time is too high and at 700K it is greater than 3000 seconds for both cases which is not the actual solution. This is because the GRI3.0 mechanism is created for a temperature of 1000K to 2500K and pressure variation from 10 Torr

(1atm = 70 Torr) to 10 atm [1]. Hence working at a temperature below 1000K is an extrapolation of the obtained data which provides inaccurate result. It can be observed that for the RMG mechanism there is a change in slope around 1000K temperature. Since there is no data from RMG for temperature below 1000K the mechanism takes the data from other libraries such as primaryThermoProperties for Thermolibraries and Klippenstein Glarborg2016 for reactionLibraries leading to change in slope.

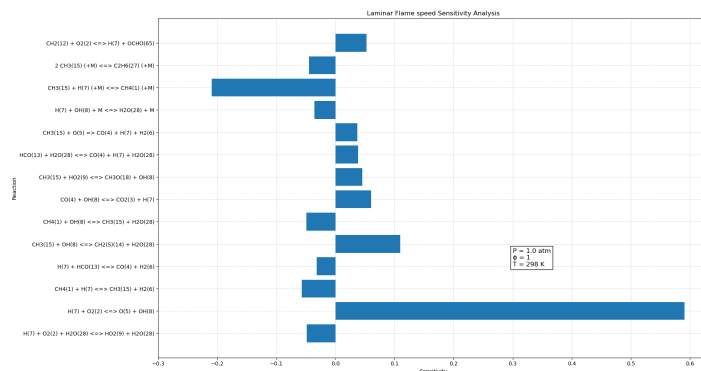


Figure 17: Laminar Flame Speed Sensitivity analysis at 1 atm Pressure by RMG mechanism

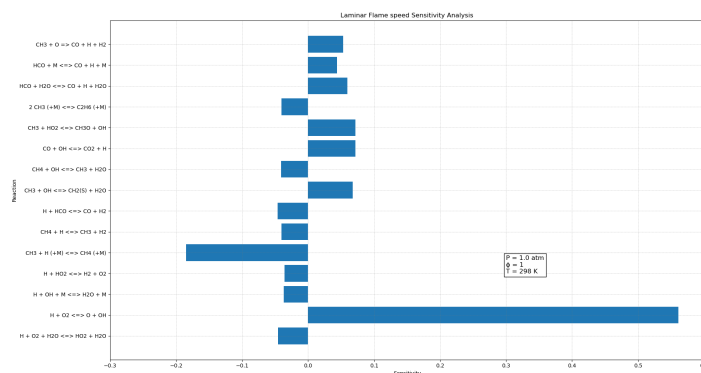


Figure 18: Laminar Flame Speed Sensitivity analysis at 1 atm Pressure by GRI3.0

From the sensitivity analysis it can be said that few of the intermediate species are not considered in the reaction and hence leading to inaccurate values and these can be avoided by explicitly adding the species to the RMG input file. From the sensitivity file it can be understood that HO_2 and CHO are not been considered in the reaction and that can lead to improper result. But considering these species we can obtain better solution.

Figure 19 and 20 shows that the main reaction path taken by both the mechanism are different leading to different answers. From RMG mechanism it can be understood that mechanism has more third body M leading to more methyl radical self reaction making that the main reaction path while for GRI3.0 mechanism it considers CH_3O formation as main reaction path. For the considered pressure the Reaction path followed by RMG mechanism might be incorrect as it requires large activation energy.

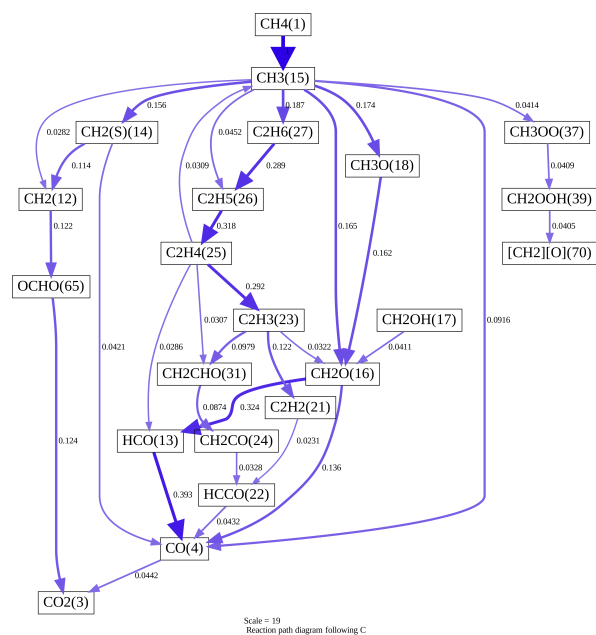


Figure 19: Reaction Path Diagram for RMG mechanism

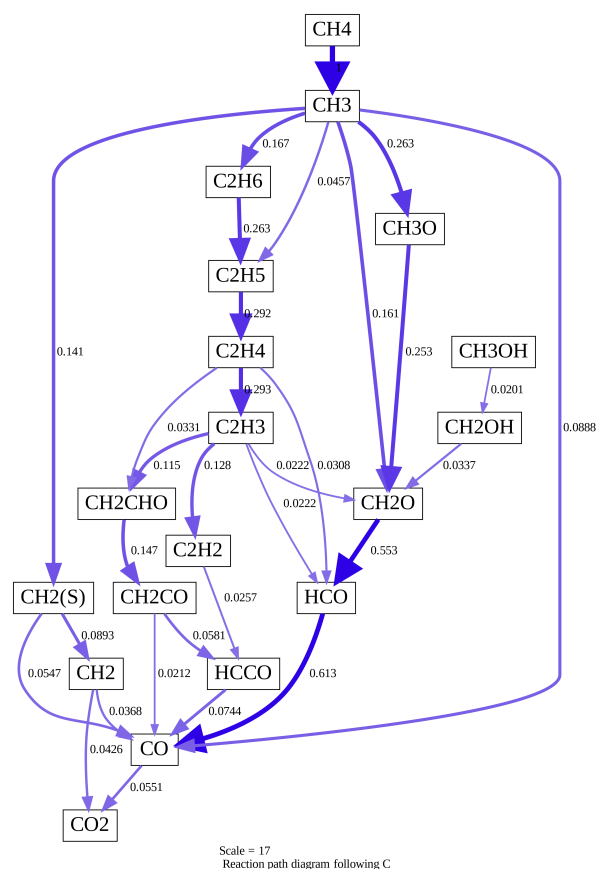


Figure 20: Reaction Path Diagram for GRI3.0 mechanism

9 Exercise 9: Reduction of methane-air mixture chemical kinetic mechanism using pyMARS

Results:

Here the reduced17 represents the reduced mechanism with 58 equations while reduced20 represents

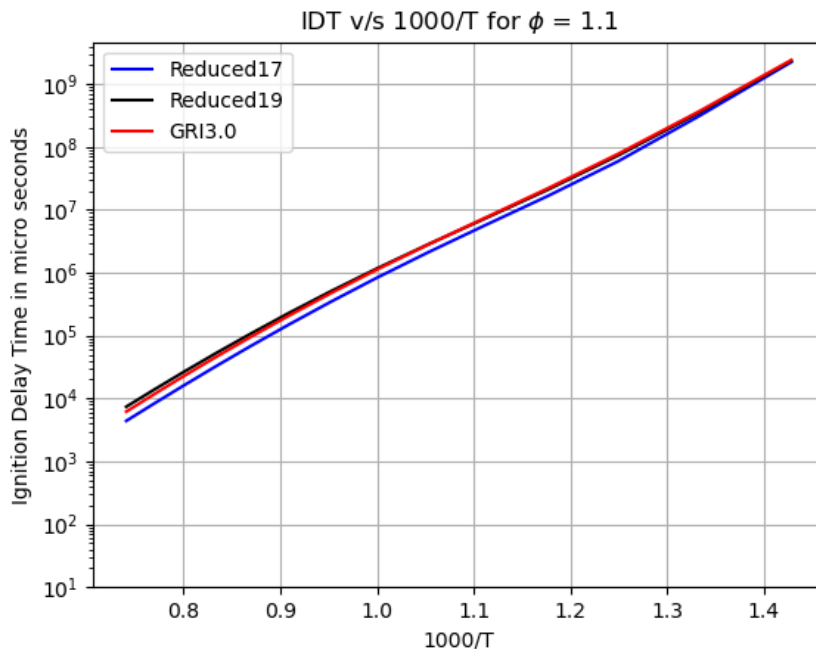


Figure 21: Comparison of Ignition delay time for reduced mechanism and GRI3.0 mechanism

the reduced mechanism with 83 equations. Here the sensitivity-type initial is used instead of greedy and can be seen that the reduced mechanism almost matches with the GRI3.0 mechanism hence for this case there is no need to use greedy condition.

From the Reaction Path diagram we can see that many of the important intermediate species are lost because of the reduction process. Hence it is important to retain some of the important intermediate species such as C_2H_5 so that more data can be saved.

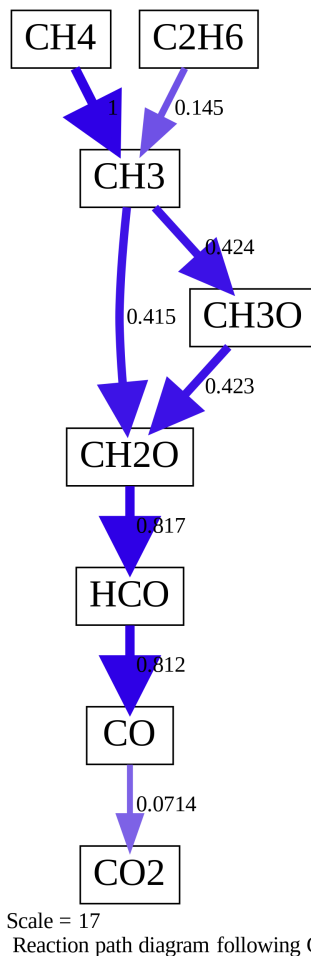


Figure 22: Reaction Path Diagram for reduced mechanism 17

References

- [1] Combustion Research Laboratory. Gri-mech 3.0. <http://combustion.berkeley.edu/gri-mech/releases.html>, 2000. Accessed: 2024-07-14.
- [2] D.P. Mishra. *FUNDAMENTALS OF COMBUSTION*. PHI Learning, 2007.
- [3] S.R. Turns. *An Introduction to Combustion: Concepts and Applications*. Number Bd. 1 in An Introduction to Combustion: Concepts and Applications. McGraw-Hill, 2000.
- [4] J. Warnatz, U. Maas, and R.W. Dibble. *Combustion: Physical and Chemical Fundamentals, Modeling and Simulation, Experiments, Pollutant Formation*. Springer Berlin Heidelberg, 2012.